PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

DISASTER PREPAREDENESS AND RESPONSE EDUCATION SYSTEM FOR SCHOOLS AND COLLEGES

A PROJECT REPORT

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BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING, CYBER SECURITY

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

PRESIDENCY UNIVERSITY

BENGALURU

APRIL 2026

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "**DISASTER PREPAREDNESS AND RESPONSE EDUCATION SYSTEM FOR SCHOOLS AND COLLEGES"** is a Bonafide work of "Dhanya D T (20221CCS0115), C Niharika (20221CCS0113) and Bhavani (20221CCS0128)" who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE ENGINEERING, CYBER SECURITY during 2025-26.

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DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE ENGINEERING, CYBER SECURITY at Presidency University, Bengaluru, named Dhanya D T, C Niharika, Bhavani, hereby declare that the project work titled **"Disaster Preparedness and Response Education System for Schools and Colleges"** has been independently carried out by us and submitted in partial fulfilment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING (CYBER SECURITY) during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

The aim of our project “Disaster Preparedness and Response Education System for Schools and Colleges” is to improve public safety by providing information, guidance, and disaster response strategies keeping in mind the various natural and human-induced calamities. The Disaster Preparedness and Response Education System focus on improving preparedness, reduce the effect of disasters, and ensure effective disaster response through a centralized digital platform. The digital platform developed is used to bring awareness and guidance on the safety procedures that has to be followed before, during, and after the disaster.

The information on various types of disasters such as earthquake, floods, cyclones, fire, and stampedes is provided by the system. Each disaster provides information about how it occurs, their effects and the safety procedures that has to be followed. This allows the users to understand the situation and take appropriate measures during the calamities. The aim of these modules is to give the correct detailed information about the disasters that occur.

The main benefits of the Disaster Preparedness and Response Education System include providing early warnings, safety awareness, and the ability to respond to the disasters in a proper manner. Providing timely information enables users to reduce the damage, loss of life and improve the effectiveness of the disaster response.

Eary warnings support, safety awareness and quick response to the emergency situations are some of the major benefits provided by the system. The system provides users with right information at the right time, which helps reduce the level of damage and save more lives.

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ABBREVIATIONS

Abbreviation	Full Form
SDG	Sustainable Development Goal
IoT	Internet of Things
IoTWF	Internet of Things World Forum
SDLC	Software Development Life Cycle
API	Application Programming Interface
GIS	Geographic Information System
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
UI	User Interface
UX	User Experience
UAT	User Acceptance Testing
PESTLE	Political, Economic, Social, Technological, Legal, Environmental
IFRC	International Federation of Red Cross and Red Crescent Societies
UNDRR	United Nations Office for Disaster Risk Reduction
FEMA	Federal Emergency Management Agency
NOAA	National Oceanic and Atmospheric Administration
NHC	National Hurricane Center

CHAPTER 1

INTRODUCTION

1.1 Background

In recent times, various reasons such as climate changes, urbanization, and increase in the population of the human beings contribute to increase in the number and severity of the disasters. For examples disasters like earthquake, floods, cyclones, fire and stampedes cause huge damage to the environment, properties and lives. Lack of preparedness, the response mechanism and lack of awareness among the people are some of the reasons due to which huge damages occur during the disasters.

In the past, the focus of disaster management was mainly based on the response mechanism after the occurrence of the disaster. However, in the recent times the focus is shifted on the importance of the preparedness. With improvement in technology, the need for disaster preparedness and management systems has increased. The main role of these systems is to improve public safety, educate them and help respond to the disaster in right manner.

1.2 Statistics of project

Disaster has become one of the critical issues in today's world due to the increase in number and severity of disaster. Collection and analyzation of statistical data is important for disaster preparedness and management systems.

The reports of global organizations like United Nations and the World Bank states that more than 200-300 million people are affected by disaster each year across the world. As per the reports of the National Oceanic and Administration (NOAA), thousands of deaths and economic losses in billions of dollars are caused by natural disasters each year. National Hurricane Center(NHC) suggests that among all the types of disasters, floods are the major ones taking up around 40% of the total number of disasters.

In India, the situation is critical due to the geographic and climatic conditions:

- More then 60% of India's landmass is vulnerable to earthquakes. The intensity levels are different at different locations.
- During the monsoon season, around 12% of the land area is flood-prone.
- 5,700 km of co
- Coastline is vulnerable to cyclones, due to which millions of people living across the coastal regions are affected.
- Electrical faults, industrial defects and human negligence are some of the reasons for fire accidents.
- Stampedes are mostly seen during religious gatherings, public events, and crowded places which leads to unavoidable situations.

As per the recent reports of researchers in the domain of disaster management, it is stated that more than 70% of the total damage caused by disasters is due to the lack of awareness. It is also stated that the implementation of the right kind of disaster preparedness mechanisms can reduce the loss of life up to 30-50%.

As per the recent trends in the implementation of modern technologies in the domain of disaster management, it is stated that the implementation of systems like the one which can provide real-time information can reduce the degree of panic among the general public. However, such systems are either not implemented or not easily accessible.

As per the statistics provided in the reports of global organizations like the United Nation and the World Bank, it is clearly stated that the implementation of the right kind of Disaster Preparedness and Management System can be useful in minimizing the damage caused by disasters.

1.3 Prior existing technologies

Disaster management has been significant changes over the past few years with the incorporation of technology. Several technologies and systems have been introduced in the field of disaster management. These systems help in predicting and managing disasters. Understanding the existing technologies in the domain is necessary to know the advantages and disadvantages of the systems and to create more efficient systems.

1.3.1 Early Warning Systems

Early warning systems are used to predict and detect disasters like cyclones, floods, and earthquakes. These systems use satellite images and weather forecasting models to predict the changes in the environment. The authorities and the public are alerted about the disasters.

Although the early warning systems are highly efficient in predicting disasters, they may face difficulties like:

- Lack of access to the public in rural areas.
- Limited detailed and clear guidance for the public.
- Delays in communication in specific circumstances.

1.3.2 Mobile Alert and Notification Systems

Mobile-based alert systems, which are often used by governments, send alerts regarding disaster in the form of SMS or mobile apps. These alerts basically provide information regarding risks and safety measures.

Advantages:

- Information can be disseminated in a quick manner
- Information can reach a large number of people because everyone uses a mobile device.

Limitations:

- Information if often in a short form and lacks adequate information.
- Alerts are often dependent on network availability.
- Alerts lack interactivity with the user.

1.3.3 Geographic Information Systems (GIS)

GIS technology helps in mapping disaster-prone areas and to analyze geographical information. It also helps in identifying high-risk zones, planning evacuation routes, and managing resources effectively.

Advantages:

- Accurate information can be obtained
- Helps in planning and decision making

Limitations:

- Requires adequate technical skills
- Information obtained from this technology is complex for the general public
- Not easily accessible to public

1.3.4 Disaster Management Portals and Websites

There are several organizations that offer information regarding disasters on various websites and portals.

Advantages:

- Information is centralized
- Information regarding disasters can be easily accessed

Limitations:

- Information is mostly static
- Information might not always be updated in real time
- Not user-friendly for all types of users

1.3.5 Emergency Response Systems

Emergency response systems include services like fire departments, ambulance services, and disaster response forces. These systems play a crucial role during and after disasters by providing rescue and relief operations.

Advantages:

- Immediate assistance during emergencies.
- These services are handled by professionals.

Limitations:

- These services play a crucial role in helping during disasters
- Delay in reaching affected areas
- Limited public awareness regarding these services

1.4 Proposed approach

The proposed "Disaster Preparedness and Management System" is envisioned as a comprehensive and user-friendly web-based system aiming to promote disaster awareness preparedness and response. The system will integrate various disaster management modules into a single centralized system, facilitating users to quickly and efficiently access disaster information in emergency situations.

Centralized Disaster Information System

The main aim of the proposed Disaster Preparedness and Management System is to develop a centralized system where users can find all disaster information at one place instead of relying on various sources. The information will be organized in a semantic manner for various types of disasters, including earthquakes, floods, cyclones, fires, and stampedes.

Each disaster module includes:

- Causes and risk factors
- Preventive measures
- Step-by-step actions before, during, and after the disaster

This structured approach helps users to clearly understand how to respond in different situations by reducing confusion and panic.

Disaster-Specific Modules

The Disaster Preparedness and Management System will be divided into various modules for different types of disaster:

- **Earthquake Module:** It provides safety measures that includes drop, cover, and hold techniques, along with preparedness tips.

- **Flood Module:** Includes evacuation guidelines, water safety tips, and emergency planning.
- **Cyclone Module:** Covers early warnings, shelter planning, and post-cyclone recovery steps.
- **Fire Module:** Provides fire prevention techniques, emergency exits, and first-aid measures.
- **Stampede Module:** Focuses on crowd management awareness and safety practices in public gatherings.

These modules ensure that users receive detailed and relevant information tailored to each disaster type.

Real-Time Alert and Notification System

One of the key components of the proposed system is the real-time alert system. The system has been designed to provide users with timely alerts concerning disasters. This enables users to take preventive measures in advance.

One of the key features of the proposed system is the real-time alert mechanism. The system is designed to deliver timely notifications about potential disasters, helping users take preventive actions in advance.

Features include:

- Instant alerts for disaster warnings
- Emergency notifications with safety instructions
- Updates related to ongoing disaster situations

This feature plays a crucial role in minimizing risks and improving response time.

Technology Stack and Architecture

The system is built using modern web technologies to ensure efficiency and scalability:

- **Frontend:** Developed using React, providing a dynamic and responsive user interface.

- **Backend:** Built using Node.js, enabling fast data processing and server-side operations.
- **Database (optional):** Can be used to store user data, alerts, and logs for future analysis.

The architecture follows a modular design, allowing easy updates and scalability without affecting existing functionalities.

1.5 Objectives

The primary aim of the Disaster Preparedness and Management System is to ensure the development of an efficient and user-friendly system that enhances awareness, preparedness, and response to various disaster situations. The system aims to reduce risk, mitigate damage, and ensure public safety through the provision of accurate information

Main objectives of this project are:

- **To Develop a Centralized Disaster Management Platform:** The primary aim is to ensure the development of an integrated system that consolidates information regarding various forms of disaster. This will ensure that users have access to all the information they require without the need to seek information from other sources.
- **To create awareness about Disasters:** The system aims to create awareness among users regarding various forms of disaster, including earthquakes, floods, cyclones, fire and stampedes. It provides information regarding causes ,impacts, and prevention.
- **To provide Step-by-Step Guidelines:** Another important objective is to guide users on what actions to take:
- Before a disaster
- During a disaster
- After a disaster
- These structured instructions help users respond effectively and reduce confusion during emergencies.
- **To Implement Real-Time Alerts and Notification:** The system is designed to deliver timely alerts and warnings related to disasters. This helps users take preventive actions in advance and reduces the impact of unexpected situations.

- **To Design a User-Friendly Interface:** The project focuses on creating a simple, intuitive, and easy-to-use interface so that users of all age groups and technical backgrounds can access information without difficulty.
- **To Reduce Loss of Life and Property:** One of the major objectives is to minimize damage caused by disasters by improving preparedness and awareness, thereby contributing to public safety.

1.6 SDGs

This project aligns with several United Nations Sustainable Development Goals (SDGs) by promoting safety, resilience, and the use of technology for societal benefit. Disaster significantly affect human life, infrastructure, and economic stability, and this project contributed to reducing these impacts through awareness and preparedness.

Fig 1.1 Sustainable Development Goals

SDG 3: Good Health and Well-being

This project supports SDG 3 by helping to reduce injuries, fatalities, and health risks during disasters. By providing timely information, safety guidelines, and emergency measures, the system ensures that individuals can take appropriate actions to protect themselves and others. For example, during disasters such as earthquakes or fires, immediate knowledge of safety procedures can prevent panic and reduce physical harm. The system also promotes mental well-being by reducing fear and confusion through clear instructions.

SDG 9: Industry, Innovation, and Infrastructure

The project contributes to SDG 9 by utilizing modern technologies such as React and Node.js to develop an innovative and scalable disaster management system. It demonstrates how digital solutions can improve infrastructure related to emergency management.

By integrating real-time alerts, structured information, and user-friendly interfaces, the system enhances technological innovation in disaster preparedness and supports the development of smart and resilient infrastructure.

SDG 11: Sustainable Cities and Communities

SDG 11 focuses on making cities and communities safe, resilient, and sustainable. This project directly contributes to this goal by improving disaster readiness among citizens.

Urban areas are highly vulnerable to disasters such as fires, floods, and stampedes due to high population density. The system helps communities become more prepared by providing guidance on evacuation, safety measures, and risk reduction strategies.

SDG 13: Climate Action

Climate change has increased the frequency and intensity of disasters such as floods and cyclones. This project supports SDG 13 by promoting awareness and preparedness for climate-related disasters.

The system educates users about the impact of climate change and encourages proactive measures to reduce risks. By providing early alerts and safety guidelines, it helps communities adapt to changing environmental conditions.

1.7 Overview of Project Report

The project report titled “Disaster Preparedness and Management System” is organized into multiple chapters, each focusing on a specific aspect of the system development, design, and evaluation. The structure of the report ensures a clear and systematic presentation of the project from introduction to conclusion.

Chapter 1 provides an overview of disaster management and importance of preparedness. It outlines the objectives of the project, its alignment with Sustainable Development Goals (SDGs), and a brief description of the entire report structure.

Chapter 2 reviews existing research and technologies related to disaster management systems. It analyses various approaches such as early warning systems, mobile alert systems, and AI-based prediction models. The chapter identifies the limitations of current systems and highlights the need for an improved, integrated solution.

Chapter 3 explains the approach used to design and develop the Disaster Preparedness and Management System. It includes system planning, requirement gathering, and the overall workflow of the system. The methodology focuses on how different disaster modules and alert systems are integrated to provide accurate and timely information.

Chapter 4 describes the planning and management aspects of the project. It includes the project timeline (Gantt chart), risk analysis, and resource management. It explains how the project was organized and executed within a specific time frame.

Chapter 5 presents the system requirements and design architecture. It includes hardware and software requirements, block diagrams, system flowcharts, and module descriptions. The chapter explains how the system is structured and how different components interact with each other.

Chapter 6 focuses on the development of the system using technologies such as React and Node.js. It explains how the frontend and backend are implemented and how the system functions in real-time.

Chapter 7 discusses the testing strategies used to ensure the system works correctly. It includes different types of testing such as functional testing, performance testing, and integration testing. The results and insights obtained from testing are also presented.

Chapter 8 highlights the broader impact of the system. It discusses how the system addresses social needs, follows legal guidelines, ensures ethical use of technology, promotes sustainability, and maintains safety standards.

Chapter 9 summarizes the overall project, including its achievements and outcomes. It also suggests possible future enhancements such as adding advanced technologies, improving features, and expanding the system for wider use.

CHAPTER 2

LITERATURE REVIEW

Disaster preparedness and response systems have been widely studied to reduce the impact of natural and man-made disasters. Various organizations and researchers have proposed different approaches such as awareness programs, simulation-based training, web-based platforms, and gamification techniques to improve disaster education.

This chapter reviews relevant literature related to disaster management systems, identifies key methodologies, and highlights research gaps addressed by the proposed system.

Review of Existing Literature

United Nations Office for Disaster Risk Reduction (UNDRR) (2023) [1] The United Nations Office for Disaster Risk Reduction (UNDRR) highlights the importance of public awareness and preparedness in minimizing disaster risks. The report emphasizes that effective disaster education can significantly reduce casualties and damage. However, the study mainly focuses on policy frameworks and lacks practical, interactive training systems for individuals.

Federal Emergency Management Agency (FEMA) (2022) [2] The Federal Emergency Management Agency provides structured disaster preparedness training programs that include drills and simulations. These programs improve emergency response skills. However, they are not easily accessible to all users and lack continuous performance evaluation.

S. Lee, J. Park, and H. Kim (2020) [6] Lee et al. developed interactive learning platforms for disaster awareness. Their study shows that simulation-based learning improves engagement and understanding. However, the system lacks proper evaluation metrics.

Sharma et al. [1] proposed a disaster prediction system using machine learning techniques to analyse environmental data such as rainfall, temperature, and seismic activity. Their model improved prediction accuracy and enabled early warning generation. However, the system mainly focused on prediction and lacked user-level guidance for disaster response.

R. Kumar and P. Singh [2] developed a flood monitoring system using IoT sensors to measure water levels in real time. The collected data was transmitted to a cloud platform for analysis and alert generation. The system provided real-time updates, but its effectiveness depended heavily on sensor deployment and network availability.

Reddy et al. [3] studied the use of Geographic Information Systems (GIS) in disaster management. Their work highlighted how GIS can map disaster-prone areas and assist in evacuation planning. While GIS improves decision-making, it requires technical expertise and is not easily accessible to common users.

Patel and Mehta [4] implemented a mobile-based alert system for disaster notifications. The system used SMS and mobile applications to send alerts to users. Although it ensured quick communication, the alerts were limited in detail and lacked interactive guidance.

Gupta et al. [5] explored the use of Artificial Intelligence and Machine Learning in disaster prediction. Their system analysed large datasets to identify patterns and predict disasters such as cyclones and floods. Despite high accuracy, the system required large datasets and computational resources.

Summary of Findings and Research Gaps

The review of existing disaster management systems and technologies highlights that significant progress has been made in improving disaster prediction, monitoring, and response mechanisms. Technologies such as Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT), Geographic Information Systems (GIS), and cloud computing have enhanced the accuracy of disaster prediction and enabled real-time monitoring of environmental conditions. Mobile applications and alert systems have improved communication by providing quick notifications to users, while GIS has supported effective planning and decision-making through mapping of disaster-prone areas. These advancements indicate a shift from traditional manual approaches to modern technology-driven disaster management systems.

However, despite these improvements, several research gaps still exist. Most existing systems focus on specific aspects such as prediction or monitoring and lack a unified and integrated platform that covers multiple disaster types. There is also limited emphasis on disaster preparedness and user awareness, as many systems primarily concentrate on post-disaster response rather than preventive measures. Additionally, advanced technologies

like GIS and AI are often complex and not easily accessible to common users, reducing their practical usability. Many systems provide alerts but fail to offer clear, step-by-step guidance on actions to be taken before, during, and after disasters. Furthermore, these systems are highly dependent on infrastructure such as internet connectivity and power supply, which may not be reliable during emergency situations. High implementation costs and lack of interactive or personalized features also limit their widespread adoption.

These limitations highlight the need for a centralized, user-friendly, and comprehensive system that focuses on preparedness, awareness, and effective response. The proposed Disaster Preparedness and Management System aims to address these gaps by integrating multiple functionalities into a single platform, providing clear guidance to users, and improving accessibility. By combining technological advancements with practical usability, the system seeks to enhance disaster readiness and reduce the overall impact of disasters.

Summary of Literatures reviewed

Reference	**Focus Area**	**Key Findings**	**Accuracy/ Performance**	**Limitations**
UNDRR (2023) [1]	Disaster awareness and policies	Awareness reduces disaster impact and casualties	High at policy level	No practical or interactive training
FEMA (2022) [2]	Disaster training programs	Simulation and drills improve response skills	Effective in controlled training	Limited accessibility, no continuous evaluation

Lee et al. (2020) [6]	Simulation-based learning	Interactive platforms improve engagement and understanding	Moderate to high	No proper performance evaluation metrics
Sharma et al. [1]	Disaster prediction using ML	Predicts disasters using environmental data	High prediction accuracy	No user-level response guidance
Kumar & Singh [2]	IoT-based flood monitoring	Real-time water level monitoring and alerts	Accurate with sensors	Depends on network& sensor deployment
Reddy et al. [3]	GIS in disaster management	Maps disaster-prone areas and supports planning	High spatial accuracy	Requires technical expertise
Patel & Mehta [4]	Mobile alert system	Provides quick disaster notifications via SMS/apps	Fast communication	Limited details, no interactivity
Gupta et al. [5]	AI/ML-based prediction	Identifies patterns for disaster prediction	High accuracy	Requires large datasets & high computation

CHAPTER 3

METHODOLOGY

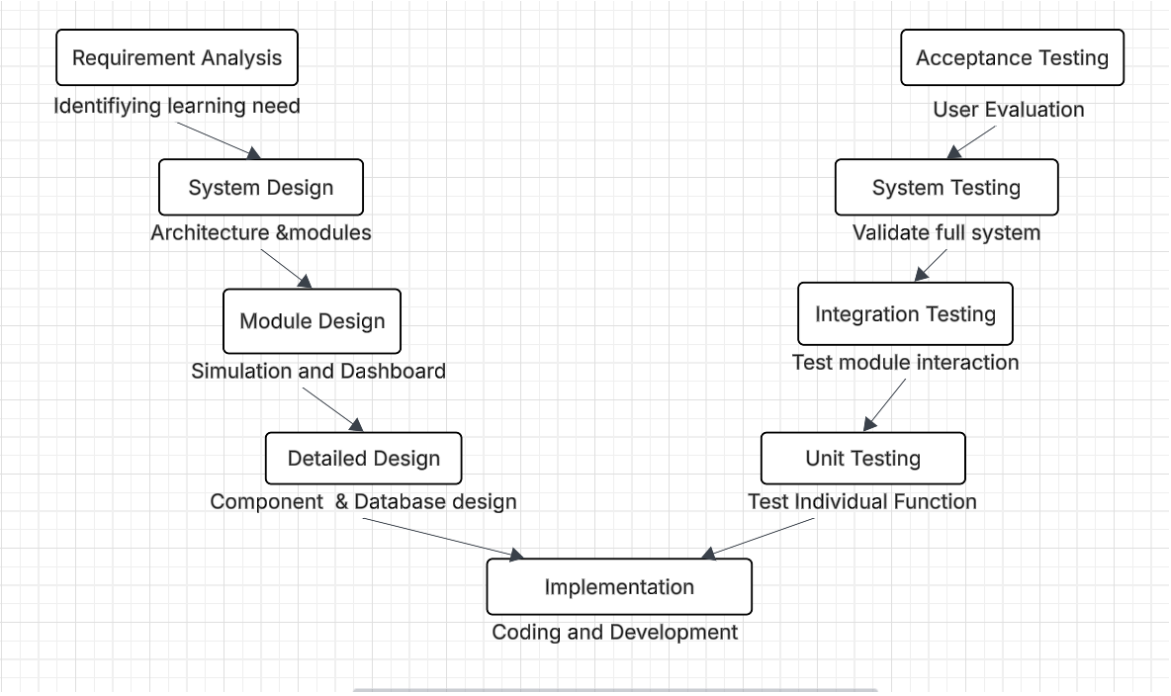

Fig 3.1 V-Model of SDLC Lifecycle

The development of the Disaster Preparedness and Management System is carried out using the V-Model (Verification and Validation Model), which is a structured and systematic approach derived from the Software Development Life Cycle (SDLC). The V-Model is chosen for this project because it ensures that each phase of development is associated with a corresponding testing phase, thereby improving the overall quality, reliability, and accuracy of the system.

The V-Model consists of two major phases, namely the Verification phase and the Validation phase. The verification phase focuses on system design and development, whereas the validation phase focuses on testing and evaluation. Each stage in the development process is validated through corresponding testing activities, ensuring that errors are identified and corrected at an early stage.

Project Approach

The development of the Disaster Preparedness and Response Education System follows the V-Model methodology, which ensures a systematic and structured approach by linking each development phase with a corresponding testing phase. This approach emphasizes both verification and validation, making it suitable for developing a reliable and user-centric system.

In the initial stages of the V-Model, the project begins with requirement analysis, where the system objectives and user needs are clearly defined. This is followed by system design and module design, where the architecture of the application is planned, including frontend, backend, and database components. The system is then implemented using technologies such as React.js, Node.js, and MongoDB.

On the validation side of the V-Model, each development phase is tested through corresponding testing stages. Unit testing is performed for individual components such as login and simulation modules, while integration testing ensures proper communication between system components. System testing validates the overall functionality of the application, and acceptance testing ensures that the system meets user requirements.

This approach ensures early detection of errors, improves system quality, and guarantees that the final product is reliable and effective for disaster preparedness training.

Idea Generation and Requirement Gathering

In the V-Model, idea generation is closely associated with the requirement analysis phase, where the need for the system is identified. The idea for this project was generated by analysing the challenges faced during disaster situations, particularly the lack of awareness and preparedness among individuals.

It was observed that existing disaster education methods are mostly theoretical and do not prepare users for real-life scenarios. People often panic during emergencies due to insufficient practical training. Additionally, existing digital platforms mainly provide information and alerts but lack interactive learning and performance evaluation features.

These observations led to the idea of developing an interactive system that combines simulation-based learning with real-time feedback and performance tracking. The concept

focuses on enabling users to experience disaster scenarios virtually and make decisions, thereby improving their response skills. Thus, the idea generation phase forms the foundation for defining system requirements in the V-Model.

Requirement gathering is a critical part of the requirement analysis phase in the V-Model, where all system needs are identified and documented before proceeding to design and development.

The requirements for this project were collected through literature review, analysis of existing technologies, and understanding user needs. The system is primarily designed for students in schools and colleges, with the goal of improving disaster awareness and preparedness.

The functional requirements include features such as user authentication, access to disaster-specific learning modules, participation in simulation-based drills, and performance tracking through a dashboard. The system also includes an admin module for managing content and monitoring user performance.

The non-functional requirements focus on system quality attributes such as performance, scalability, security, and usability. The system must provide fast response times, handle multiple users efficiently, ensure data security, and offer a user-friendly interface.

By clearly defining these requirements at the initial stage, the V-Model ensures that the system is developed in a structured manner and that each requirement is validated during the testing phases, ultimately leading to a reliable and effective disaster preparedness system.

Technology Stack

Node.js & Express.js: Used to build backend APIs for handling modules, drills, scores, and user authentication.

MongoDB (Atlas): Used as a database to store user data, scores, and module progress.

React.js: Used to create a dynamic and responsive user interface with reusable components.

HTML, CSS, JavaScript: Used for structuring, styling, and adding interactivity to the application.

Future Enhancements

The Disaster Preparedness and Management System can be further improved by incorporating advanced features such as integration with real-time disaster and weather APIs to provide live updates, development of a mobile application for better accessibility, and implementation of AI-based prediction models to enhance disaster forecasting. Additional features like multilingual support can make the system accessible to a wider audience, while voice assistance can help users understand safety instructions more easily. The system can also be enhanced with GPS-based location tracking to provide area-specific alerts and integrated with government disaster management platforms for more accurate and reliable information. These improvements will make the system more intelligent, user-friendly, and effective in handling real-world disaster situations.

CHAPTER 4

PROJECT MANAGEMENT

4.1 Project timeline (Gantt Chart)

The "Disaster Preparedness and Response Education System for Schools and Colleges" was done over a period of six months, from January to April 2026.

The estimated timeline was discussed and developed during planning phase. A detailed Gantt chart was developed to track progress constantly.

Every month focused on:

- January 2026: Discuss the requirements gathering and in-depth literature review to get familiar with the challenges faced by traditional systems.
- February 2026: Setting up the technologies and preparing the modules.
- March 2026: Developing the backend and creating first phase of UI.
- April 2026: Integrating frontend, backend with the database.

Connecting every module, expand FAQ section, testing chatbot features.

Final system tests, bug fixing along with documentation, project submission and presentation.

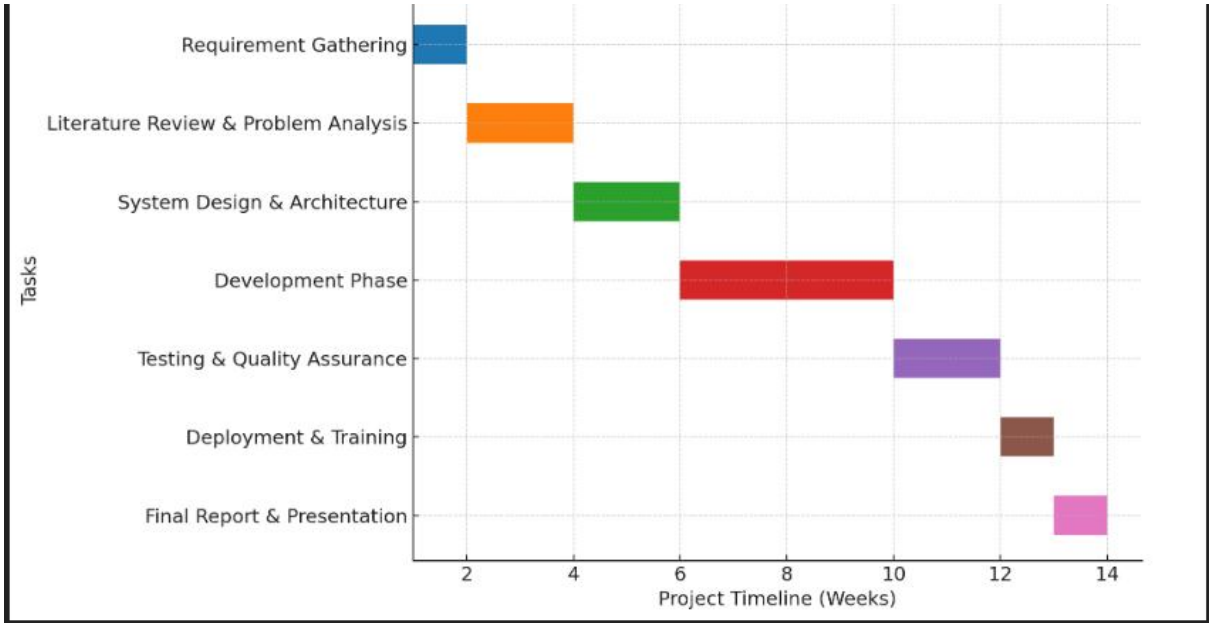

Fig 4.1 Gantt Chart

4.2 Risk analysis

Risk analysis is an important aspect of project management that identifies potential risks and their impact on the project. In this project, risks are analysed using the PESTLE approach, which considers political, economic, social, technological, legal, and environmental factors.

Technological risks include system failures, bugs, or integration issues between frontend and backend components. These risks are mitigated through proper testing and debugging. Social risks involve lack of user engagement or awareness, which can be addressed by designing an interactive and user-friendly system. Economic risks are minimal as the project uses open-source technologies. Legal risks include data privacy concerns, which are handled through secure authentication and data protection mechanisms.

By identifying these risks early, appropriate mitigation strategies are implemented to ensure smooth project execution.

CHAPTER 5

ANALYSIS AND DESIGN

5.1 Requirements

5.1.1 Hardware Requirements

Components	Specification
Processor	Intel Core i5
RAM	Minimum of 4GB
Hard Disk	256 GB SSD or higher
Display	1366 x 768 resolution or higher
Network	Internet connectivity (for API and database testing)

5.1.2 Software Requirements

Components	Specification
Operating System	Windows10/11
Backend Framework	Node.js
Frontend Technologies	HTML, CSS, JavaScript
Database	MongoDB
Developments Environment	Visual Studio Code
Version Control	GitHub

5.2 Block Diagram

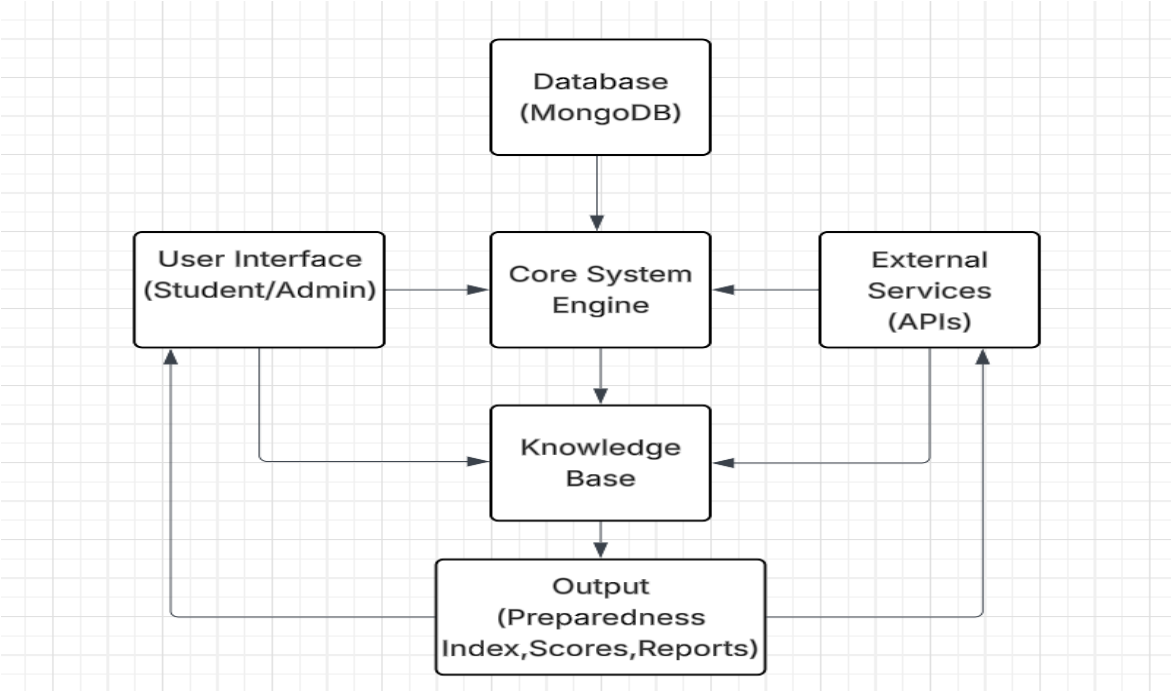

Fig 5.2 System Architecture Design

5.3 System Flow Chart

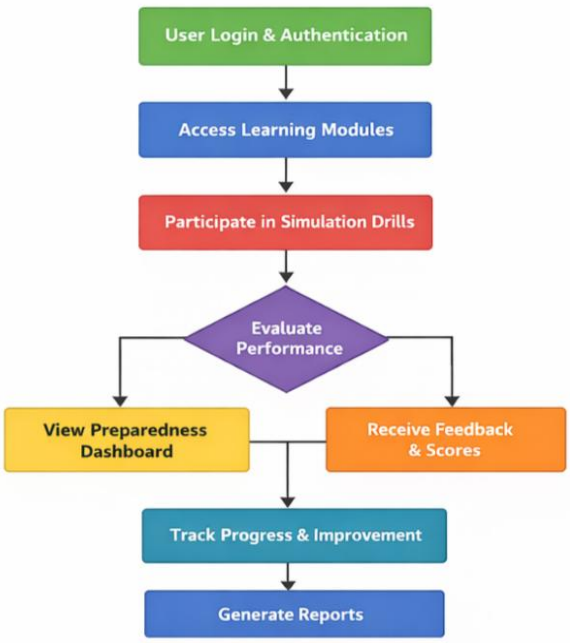

Fig 5.3 User Interaction Flow

5.4 Choosing devices

The system is designed as a web-based application, it supports devices such as desktops, laptops, tablets, and smartphones. The selection of devices is based on accessibility and usability, ensuring that users can access the system from different platforms without the need for specialized equipment. This enhances the reach and effectiveness of the system.

Devices Used:

- Personal computer or laptop with Intel Core i5 processor.
- 4 GB RAM that handles semantic models.
- A 256 GB SSD for storing the knowledge base, embeddings and logs.
- Standard display for UI development and testing.

These devices are sufficient to run the backend functions and database handling efficiently.

5.5 Designing System Units

The project is divided into modular components where each component performs a specific task. They work independently but integrate together to solve user problems accurately.

Main system units:

1. User Interface Unit

Provides interaction between the user and the system through a graphical interface. Allows users to access modules, simulations, and dashboards easily.

2. Learning Module Unit

Contains structured educational content for different disasters. Helps users understand safety measures and precautions.

3. Simulation Unit

Provides real-life disaster scenarios in the form of drills. Evaluates user decisions and gives feedback.

4. Backend Processing Unit

Handles application logic and processes user requests. Ensures communication between frontend and database.

5. Evaluation Unit

Calculates user performance using the Preparedness Index. Helps identify strengths and areas for improvement.

6. Admin Unit

Allows administrators to manage system content and users. Monitors performance and updates modules.

7. Communication Unit

Manages data exchange using APIs. Ensures smooth interaction between system components.

5.6 Standards

A set of rules were followed to ensure that the system works well and successfully establishes coding, organizing knowledge and design user experience.

Rules and Practices followed:

1. Coding Standards

Maintained consistent naming conventions and proper code structure.
Used indentation and modular coding for readability and maintenance.

2. Web Development Standards

Followed HTML5, CSS3, and JavaScript best practices.
Ensured responsive design and cross-browser compatibility.

3. API Standards

Implemented RESTful APIs for communication between components.
Used standard HTTP methods for efficient data handling.

4. Security Standards

Applied authentication and authorization mechanisms.

Ensured protection of user data and prevented unauthorized access.

5. Testing Standards

Performed unit, integration, and system testing.

Ensured system reliability before deployment.

5.7 Mapping with IoTWF Reference Model Layers

Although IoTWF is generally used for IoT systems, this project follows same layered approach to map how data flows across different system components.

Layers	**Equivalent in Project**	**Description**
Perception Layer	User Interface (Student & Admin)	Captures user inputs such as login details, module selection and response to disaster simulation drills
Connectivity Layer	Internet, HTTP Protocol, REST APIs	Enables communication between frontend, backend and database for smooth data transfer.
Edge Computing Layer	Backend Server	Processes user requests, evaluates simulation responses, and handles application logic.
Data Accumulation Layer	MongoDB	Stores user data, learning progress, drill scores, and preparedness index.
Data Abstraction Layer	API &Data Processing Layer	Converts raw data into meaningful insights such as performance metrics and feedback.
Application Layer	Web Application (Modules &Dashboard)	Provides disaster education content, simulation interface, and performance visualization.

Collaboration & Process Layer	User-system interaction & admin control	Supports learning, monitoring, feedback, and management of system activities.

5.8 Domain Model Specification

The project surrounds around key entities and their relationships which helps to organize all information smoothly.

Primary entities:

1. User Entity

- Represents students or users who can access the system
- Interacts with learning modules and simulation drills

2. Learning module entity

- Contains disaster-related educational content.
- Includes modules like earthquake, flood, cyclone, fire, stampede.
- Provides instructions, precautions and guidelines

3. Simulation Drill Entity

- Represents real-life disaster scenarios.
- Includes question, options, and correct responses.
- Evaluates user decision-making skills.

4. Performance Record Entity

- Stores user results from simulations and modules.
- Includes scores, attempts, and completion status.
- Helps track learning progress.

5. Preparedness Index Entity

- Measures user preparedness level
- Calculated based on accuracy, consistency, and completion.
- Provides overall performance evaluation.

6. Admin Entity

- Represents system administrators.
- Manages modules, users, and system content.
- Monitors performance and system activities.

5.9 Communication Model

The communication model shows how different components exchange information during query resolution.

Flow of communication:

1.The user opens the website, and the frontend loads the interface through the browser.

2.The user enters login or registration details, which are sent to the backend server using API requests for verification.

3.After successful authentication, the user selects a disaster module, and the frontend requests module data from the backend.

4.The backend retrieves the required content from the MongoDB database and sends it back to the frontend for display.

5.During simulation drills, user responses are sent to the backend, where they are processed, evaluated, and stored in the database.

6.The processed results are returned to the frontend, and the dashboard displays scores, feedback, and preparedness index to the user.

5.10 Deployment Level

Deployment levels:

- The system is deployed as a web-based application, allowing users to access it through browsers without installing any software.
- The frontend (React.js) is hosted on a web server, providing the user interface for students and administrators.
- The backend (Node.js) is deployed on a server to handle application logic, API requests, and user authentication.

- The MongoDB database is hosted either locally or on a cloud platform to store user data, progress, and performance metrics.
- The system supports deployment on cloud platforms, enabling scalability and access for multiple users from different locations.
- Users such as students and administrators can access the system anytime using devices like laptops, tablets, or smartphones.

5.11 Functional View

The functional view provides insights on what system does rather than how it is implemented.

Main functional components:

- **User Authentication Component**: Handles user registration, login, and secure access to the system.
- **Learning Module Component**: Provides disaster-related educational content such as earthquake, flood, cyclone, fire, and stampede.
- **Simulation Component**: Presents real-life disaster scenarios and allows users to select appropriate actions.
- **Evaluation Component**: Processes user responses and calculates scores and Preparedness Index.
- **Dashboard Component**: Displays user performance, progress, and feedback in a structured format.
- **Database Management Component**: Stores and manages user data, module content, and simulation results.
- **Admin Management Component**: Allows administrators to manage modules, users, and monitor system performance.

5.12 Mapping of the IoT Deployment Levels with Functional View

Deployment Level	**Functional Component**	**Description**
Device Level	User Authentication & Input Component	

		Captures user input such as login details and simulation responses.
Processing Level	Backend & Evaluation	Processes suer requests, evaluates responses, and generates results,
Application Level	Learning module & Dashboard Component.	Provides disaster content, simulations, and displays performance results.

5.13 Operational View

The operational view helps us to know how the system works during the runtime.

Operational workflow:

1. User opens the disaster preparedness website through a browser.
2. User logs in or registers to access the system.
3. User selects a disaster module (earthquake, flood, cyclone, etc.).
4. System displays learning content and safety instructions.
5. User starts simulation drill and selects responses for given scenarios.
6. System evaluates responses and calculates score and preparedness index.
7. Results and feedback are displayed on the user dashboard.
8. System stores user progress and performance in the database.
9. Admin monitors user performance and manages system content.

This view helps to solve end to end functioning of the system during its actual usage.

CHAPTER 6

SIMULATION AND SOFTWARE IMPLEMENTATION

6.1 Introduction

Simulation and software implementation form the backbone of the proposed Disaster Preparedness and Response Education System. While the earlier chapters focused on design and methodology, this chapter explains how the system was practically implemented using modern software tools and frameworks. The goal is to create an interactive, scalable, and user-friendly platform that allows students and faculty to engage in disaster preparedness drills, track their progress, and receive performance feedback.

The implementation follows a three-tier architecture consisting of:

- Frontend Layer (React) – for user interaction and visualization.
- Backend Layer (Node.js) – for authentication, data processing, and API handling.
- Database Layer (MongoDB) – for storing user data, drill scores, and preparedness metrics.

Simulation drills are embedded into the platform to mimic real-life disaster scenarios. These drills are presented as interactive multiple-choice questions, where users must make quick decisions under simulated emergency conditions. The system evaluates responses and updates the Preparedness Index, which quantifies the user's disaster readiness.

6.2 Steps Involved

Step 1: Setting up the Environment

- Install Node.js and npm for backend development.
- Install MongoDB locally or configure MongoDB Atlas for cloud database.
- Set up React.js environment using Vite.
- Install required dependencies (express, mongoose, cors, dotenv).
- Configure GitHub repository for version control.

Step 2: Starting the backend Server

- Create a Node.js server (server.js) using Express.
- Define API routes for modules, drills, scores and authentication
- Connect backend with MongoDB using Mongoose.
- Start the server using the command: node server.js
- Verify API endpoints with Postman or browser requests.

Step 3: Designing Learning Modules

Each disaster is converted into a structured learning module containing:

- Introduction to the disaster
- Causes and effects
- Safety precautions (before, during, after)

These modules help students gain theoretical understanding before attempting simulations.

Steps 4: Developing Simulation Scenarios

The core implementation focuses on creating simulation-based drills. For each disaster:

- Real-life scenarios are designed
- Multiple-choice decision-based questions are provided
- Users must choose the correct action

For example, in an earthquake scenario, users are asked what action to take during shaking. This practical approach improves decision-making skills.

Step 5: Frontend Implementation (User Interface)

The frontend is developed using React.js to provide an interactive interface. It includes:

- Login and registration pages
- Module selection screen

- Simulation interface
- Dashboard displaying performance

The interface is designed to be simple and easy for students to use.

5. Backend Implementation (Logic & Processing)

The backend is implemented using Node.js, which handles:

- User authentication
- Processing simulation responses
- Calculating scores and Preparedness Index
- Managing communication between frontend and database

This ensures efficient system functionality and real-time response.

6. Database Implementation

MongoDB is used to store:

- User details
- Module progress
- Simulation results
- Preparedness scores

The database ensures that user data is securely stored and easily retrieved.

7. Evaluation Mechanism Implementation

An evaluation system is implemented to measure user performance. It calculates a Preparedness Index based on:

- Accuracy of answers
- Completion of modules
- Consistency in performance

This helps students understand their preparedness level.

8. Integration of Components

All system components, including frontend, backend, database, and simulation modules, are integrated using APIs. This ensures smooth communication and seamless functioning of the system.

9. Testing with Disaster Scenarios

The system is tested using different disaster scenarios to ensure:

- Correct functioning of simulations
- Accurate evaluation of responses
- Proper display of results

Testing ensures reliability and correctness of the system.

6.3 Software code

```
function App() {
  const saved = localStorage.getItem("user");
  return saved ? JSON.parse(saved) : null;
});

const navigateToProfileRef = useRef(null); // ← moved here, before return

const [darkMode, setDarkMode] = useState(() => {
  return localStorage.getItem("disaster_prep_theme") === "dark";
});

const toggleDarkMode = () => {
  const next = !darkMode;
  setDarkMode(next);
  localStorage.setItem("disaster_prep_theme", next ? "dark" : "light");
};

const handleLogin = (userData) => {
  setUser(userData);
  localStorage.setItem("user", JSON.stringify(userData));
};

const handleLogout = () => {
  setUser(null);
  localStorage.removeItem("user");
};

const appStyle = {
  minHeight: "100vh",
  fontFamily: "'DM Sans', sans-serif",
  background: darkMode ? "■#0F1A1A" : "linear-gradient(135deg, □#E8F5E9 0%, □#C8E6C9 100%)",
  color: darkMode ? "□#FFFFFF" : "■#000000",
  transition: "all 0.3s ease",
};

return (
  <div style={appStyle}>
    <Navbar
      user={user}
      onLogout={handleLogout}
      darkMode={darkMode}
      toggleDarkMode={toggleDarkMode}
      onNavigateProfile={(() => navigateToProfileRef.current?.())}
    />
    <Routes>
      <Route path="/" element={<Landing />} />
      <Route path="/login/:role" element={<Login onLogin={handleLogin} />} />
      <Route path="/student" element={
        <Dashboard user={user} darkMode={darkMode} onLogout={handleLogout} toggleDarkMode={toggleDarkMode} navigateToProfileRef={navigateToProfileRef} />
      } />
      <Route path="/student/dashboard" element={
        <Dashboard user={user} darkMode={darkMode} onLogout={handleLogout} toggleDarkMode={toggleDarkMode} navigateToProfileRef={navigateToProfileRef} />
      } />
      <Route path="/student/categories" element={<CategorySelect user={user} />} />
      <Route path="/student/category/:category" element={<Learn user={user} />} />
      <Route path="/student/drills" element={<DrillCategory user={user} />} />
      <Route path="/student/drill-category/:category" element={<DrillList />} />
      <Route path="/student/drill/:module" element={<Drill />} />
      <Route path="/student/result" element={<Result />} />
      <Route path="/admin" element={<Admin user={user} darkMode={darkMode} toggleDarkMode={toggleDarkMode} />} />
    </Routes>
  </div>
);
}

export default App;
```

frontend > src > services > JS api.js > ...

```javascript
const BASE = "/api";

function getToken() {
  const user = localStorage.getItem("user");
  if (!user) return null;
  return JSON.parse(user).token || null;
}

async function request(path, options = {}) {
  const token = getToken();
  const headers = { "Content-Type": "application/json" };
  if (token) headers["Authorization"] = `Bearer ${token}`;

  const res  = await fetch(`${BASE}${path}`, { headers, ...options });
  const data = await res.json();
  if (!res.ok) throw new Error(data.error || "Request failed");
  return { data };
}

// — Auth ————————————————————————————————————
export const loginUser    = (payload) =>
  request("/auth/login",    { method: "POST", body: JSON.stringify(payload) });

export const registerUser = (payload) =>
  request("/auth/register", { method: "POST", body: JSON.stringify(payload) });

// — Modules ——————————————————————————————————
export const getModules   = ()         => request("/modules");
export const addModule    = (payload)  => request("/modules",     { method: "POST",   body: JSON.stringify(payload) });
export const updateModule = (id, data) => request(`/modules/${id}`, { method: "PUT",  body: JSON.stringify(data)   });
export const deleteModule = (id)       => request(`/modules/${id}`, { method: "DELETE" });

// — Scores ——————————————————————————————————
export const saveScore     = (payload)  => request("/scores",     { method: "POST", body: JSON.stringify(payload) });
export const getScores     = ()         => request("/scores");
export const getMyScores   = ()         => request("/scores/me");
export const getUserScores = (username) => request(`/scores/${username}`);

// — Drills ——————————————————————————————————
export const getDrills           = ()         => request("/drills");
export const getDrillsByCategory = (category) => request(`/drills/category/${category}`);
export const getDrillByModule    = (module)   => request(`/drills/module/${module}`);

// — Learning ————————————————————————————————
export const getLearningByCategory = (category) => request(`/learning/category/${category}`);
export const getLearningByModule   = (module)    => request(`/learning/module/${module}`);
```

CHAPTER 7

TESTING AND EVALUATION

Testing and evaluation are critical phases in the development of the Disaster Preparedness and Response Education System for Schools and Colleges. This phase ensures that the system functions correctly, meets user requirements, and provides accurate and reliable results. Since the system involves simulation-based learning and performance evaluation, it is essential to verify that all modules operate effectively and provide meaningful feedback to users.

The testing process includes verifying individual components, checking integration between modules, and validating the overall system performance. Evaluation focuses on analysing how well the system improves disaster awareness and decision-making skills among users. This chapter presents the test points, test plan, test results, and insights obtained from system testing.

7.1 Test Points

The testing of the Disaster Preparedness and Response Education System for Schools and Colleges focuses on validating all critical functionalities to ensure that the system operates reliably and effectively in a real educational environment. The user authentication module is thoroughly tested to ensure that students and administrators can securely register, log in, and access the system without errors. This includes verifying input validation, error handling for incorrect credentials, and session management.

The learning module is tested to confirm that disaster-specific content such as earthquake, flood, cyclone, fire, and stampede is correctly displayed, properly structured, and easily accessible. Special attention is given to ensuring that safety instructions for “before, during and after” disaster situations are presented clearly, as this is essential for educational effectiveness.

The simulation module, which is the core component of the system, is rigorously tested to ensure that real-life disaster scenarios are accurately represented. Each simulation is checked for proper question display, correct handling of user responses, and smooth navigation between scenarios. The system is also tested to verify that it correctly identifies

right and wrong answers and provides appropriate feedback, helping users understand correct actions during emergencies.

The evaluation component is tested to ensure accurate calculation of scores and the Preparedness Index. This includes verifying that the system correctly considers factors such as accuracy, number of attempts, and module completion. The consistency of performance tracking across multiple sessions is also tested to ensure reliability.

The dashboard functionality is evaluated to confirm that it displays user performance, scores, and feedback in real time. It is also tested for responsiveness and clarity, ensuring that students can easily interpret their results and identify areas for improvement. Additionally, the database system is tested to ensure secure and efficient storage of user data, including login information, progress, and simulation results.

The communication between frontend and backend is tested through API calls to ensure seamless data exchange. This includes verifying that requests such as module retrieval, simulation submission, and result updates are processed correctly without delays. The system is also tested under multiple user conditions to ensure it can handle concurrent access without performance degradation.

Finally, usability testing is performed to ensure that the system is easy to use for students in schools and colleges. This includes checking navigation, interface clarity, and overall user experience. These comprehensive test points ensure that the system is reliable, accurate, and effective in delivering disaster preparedness education.

7.2 Test Plan

The test plan for the Disaster Preparedness and Response Education System is designed to ensure that all components of the system function correctly, efficiently, and reliably in a real-world educational environment. The testing process follows a structured approach based on the V-Model methodology, where each development phase is associated with a corresponding testing activity. The main objective of the test plan is to identify errors, verify system functionality, and ensure that the system meets user requirements.

The testing process begins with unit testing, where individual components of the system such as user authentication, learning modules, simulation drills, and database operations are tested independently. This helps in identifying errors at an early stage and ensures that

each module performs its intended function correctly. For example, the login module is tested for proper validation, while simulation modules are tested for correct display and response handling.

After unit testing, integration testing is performed to verify the interaction between different components of the system. This includes testing communication between the frontend (React.js), backend (Node.js), and database (MongoDB). The objective is to ensure that data flows smoothly between components, such as retrieving module content, submitting simulation responses, and updating user performance.

The next phase is system testing, where the entire application is tested as a complete system. This involves checking all functionalities together, including user navigation, simulation workflow, evaluation process, and dashboard display. System testing ensures that the application behaves as expected under normal usage conditions.

Finally, user acceptance testing (UAT) is conducted to evaluate the system from the user's perspective. Students and users interact with the system to assess its usability, effectiveness, and overall experience. Feedback collected during this phase is used to make improvements and ensure that the system meets the needs of its target audience.

The test plan also includes testing under different conditions, such as multiple user access and varying inputs, to ensure system stability and performance. Error handling and security aspects are also verified to ensure safe and reliable operation. Through this comprehensive testing strategy, the system is validated to provide accurate, efficient, and user-friendly disaster preparedness training.

7.3 Test Results

The testing of the Disaster Preparedness and Response Education System for Schools and Colleges demonstrated that the system performs reliably across all major functionalities and successfully achieves its intended objectives. The user authentication module was tested under various conditions, including valid and invalid inputs, and it consistently provided correct responses, ensuring secure access to the system. The system was able to handle multiple user registrations and logins without errors, confirming its stability for real-world usage in educational environments.

The learning modules covering disasters such as earthquake, flood, cyclone, fire, and stampede were thoroughly tested and found to be functioning effectively. All content was displayed correctly, and navigation between different sections was smooth and intuitive. The structured presentation of safety measures (before, during, and after disasters) was verified to be clear and understandable for students, ensuring effective knowledge delivery.

The simulation module, which is the core feature of the system, showed strong performance during testing. All disaster scenarios were executed correctly, and user interactions were processed without delay. The system accurately captured user responses and evaluated them in real time. The feedback mechanism worked effectively by providing immediate suggestions and correct answers, helping users understand appropriate actions during emergencies. This confirms that the system successfully delivers practical learning experiences.

The evaluation system produced consistent and accurate results. The Preparedness Index was calculated correctly based on parameters such as accuracy, number of attempts, and module completion. The system maintains consistency in performance tracking across multiple sessions, ensuring that users could monitor their progress over time. This feature proved useful in identifying user strengths and areas that require improvement.

The dashboard functionality was tested extensively and showed reliable performance. It displayed real-time updates of user scores, progress, and feedback without any delays. The visual representation of performance helped users easily interpret their results. The backend and database integration was also validated, as all user data, including simulation results and progress, were stored and retrieved accurately from MongoDB.

The system was further tested under different usage conditions, including multiple users accessing the platform simultaneously. The application maintained stable performance without significant slowdown, indicating good scalability. API communication between frontend and backend was smooth, with all requests processed correctly and efficiently.

Usability testing conducted with sample users (students) showed that the system is easy to understand and use. Users were able to navigate through modules, attempt simulations, and view results without confusion. This confirms that the system is suitable for deployment in schools and colleges.

Overall, the test results confirm that the system is reliable, efficient, and effective in delivering disaster preparedness education. The integration of simulation-based learning and performance evaluation has proven successful in enhancing user engagement and improving disaster response understanding.

7.4 Insights

The evaluation of the Disaster Preparedness and Response Education System for Schools and Colleges provide several important insights regarding its effectiveness and impact on user learning. One of the key observations is that simulation-based learning significantly enhances user engagement compared to traditional theoretical methods. Students showed greater interest and involvement when interacting with real-life disaster scenarios, as they were required to actively think and make decisions rather than passively read content.

Another important insight is that the system improves decision-making skills during emergency situations. By participating in simulation drills, users were able to understand the correct actions to be taken during disasters such as earthquakes, floods, and fires. The immediate feedback provided by the system helped users learn from their mistakes and reinforced correct responses, leading to better retention of knowledge.

The Preparedness Index proved to be an effective metric for evaluating user performance. It allowed users to track their progress over time and identify areas where improvement is needed. This personalized feedback mechanism encouraged continuous learning and motivated users to perform better in subsequent attempts.

The system also demonstrated high usability and accessibility, making it suitable for students in schools and colleges. The simple and user-friendly interface enabled users with different levels of technical knowledge to easily navigate the system and participate in learning activities. This indicates that the system can be effectively integrated into educational environments.

Additionally, the results highlight that combining education, simulation, and evaluation into a single platform provides a more comprehensive approach to disaster preparedness. Unlike existing systems that focus only on awareness or alerts, this system offers practical training along with performance tracking, making it more effective in preparing users for real-life situations.

Overall, the insights confirm that the proposed system successfully achieves its objective of improving disaster awareness and response skills. The integration of interactive learning, simulation, and performance evaluation makes it a valuable tool for disaster preparedness education in schools and college.

CHAPTER 8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILTY & SAFETY ASPECTS

The Disaster Preparedness and Response Education System for Schools and Colleges is not only a technical solution but also has significant social, legal, ethical, sustainability, and safety implications. This chapter discusses how the system contributes to society, adheres to legal standards, follows ethical practices, supports sustainability, and ensures user safety. Considering these aspects is essential to ensure that the system is responsible, reliable, and beneficial for its users.

8.1 Social Aspects

The system plays an important role in improving disaster awareness among students and educational institutions. By providing interactive learning and simulation-based training, it helps individuals understand how to respond effectively during emergencies. This contributes to building a more informed and prepared society.

Additionally, the system promotes equal access to disaster education, as it can be accessed by students from different locations through the internet. It supports community safety by encouraging responsible behaviour and preparedness. The system also helps reduce panic and confusion during disasters by training users in advance.

8.2 Legal Aspects

The system considers legal requirements related to data protection and user privacy. User information such as login details and performance data is securely stored and managed. Proper authentication mechanisms are implemented to prevent unauthorized access.

The system also ensures that the content used for disaster education follows guidelines provided by authorized organizations such as government bodies and disaster management authorities. This ensures that the information provided is accurate and legally compliant. Additionally, intellectual property rights are respected by using original or properly referenced content.

8.3 Ethical Aspects

Ethical considerations are important in ensuring that the system is used responsibly. The system is designed solely for educational purposes and does not misuse user data. It maintains transparency by providing accurate feedback and avoiding misleading information.

The system also ensures fairness by providing equal opportunities for all users to learn and improve. It avoids bias in evaluation and ensures that all users are assessed based on their performance. Furthermore, the system promotes responsible behaviour by encouraging users to follow correct safety practices during disasters.

8.4 Sustainability Aspects

The system supports sustainability by promoting digital learning, which reduces the need for printed materials and physical resources. It is designed to be scalable, allowing new disaster modules and features to be added in the future without major changes.

By improving disaster preparedness, the system contributes to reducing the long-term impact of disasters on society. It also aligns with sustainable development goals by enhancing education and community resilience. The use of web-based technologies ensures efficient resource utilization and long-term usability.

8.5 Safety Aspects

Safety is a key focus of the system, as it aims to train users to respond effectively during disasters. The system provides accurate safety instructions and guidelines for different disaster scenarios, helping users understand the correct actions to take.

The simulation-based approach ensures that users can practice decision-making in a safe virtual environment without real-world risks. The system also ensures secure handling of user data, preventing misuse and maintaining confidentiality. Overall, the system enhances both digital safety and physical safety awareness.

CHAPTER 9

CONCLUSION

The Disaster Preparedness and Response Education System for Schools and Colleges successfully addresses the critical need for effective disaster education among students. Traditional methods of disaster awareness are often limited to theoretical knowledge, which does not adequately prepare individuals for real-life emergency situations. This project overcomes these limitations by integrating interactive learning, simulation-based training, and performance evaluation into a single platform.

The system provides structured learning modules for various disasters such as earthquakes, floods, cyclones, fires, and stampedes, enabling users to understand causes, effects, and safety measures. The simulation module enhances practical understanding by allowing users to experience real-life scenarios and make decisions in a safe virtual environment. The evaluation mechanism, through the Preparedness Index, offers valuable insights into user performance and encourages continuous improvement.

The implementation using modern web technologies such as React.js, Node.js, and MongoDB ensures that the system is scalable, efficient, and user-friendly. The testing and evaluation results confirm that the system functions reliably and effectively meets its objectives. Furthermore, the system contributes to building a more aware and prepared society by improving disaster response skills among students.

Overall, the project demonstrates a successful integration of education, technology, and simulation to create a comprehensive disaster preparedness solution for educational institutions.

Future Enhancements

Although the system achieves its intended goals, there are several opportunities for further improvement and expansion.

The system can be extended into a mobile application to increase accessibility and allow students to learn and practice anytime and anywhere. Integration of real-time disaster alerts using external APIs can provide users with up-to-date information about ongoing

disasters. Artificial Intelligence (AI) can be incorporated to personalize learning experiences based on user performance and behaviour.

Advanced simulation techniques such as 3D visualization or virtual reality (VR) can be introduced to create more immersive and realistic disaster scenarios. Multilingual support can be added to make the system accessible to a wider audience across different regions. Additionally, integration with school and college management systems can help track student participation and performance more effectively.

Future enhancements may also include adding more disaster types, improving analytics for better insights, and incorporating collaborative features where users can participate in group-based simulations. These improvements will further strengthen the system and increase its impact on disaster preparedness education.

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Review

A Systematic Review of Disaster Management Systems: Approaches, Challenges, and Future Directions

Saad Mazhar Khan [1], Imran Shafi [1], Wasi Haider Butt [1], Isabel de la Torre Diez [2,*], Miguel Angel López Flores [3,4,5], Juan Castanedo Galán [3,6,7] and Imran Ashraf [8,*]

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[8] Department of Information and Communication Engineering, Yeungnam University, Gyeongsan 38541, Republic of Korea
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Abstract: Disaster management is a critical area that requires efficient methods and techniques to address various challenges. This comprehensive assessment offers an in-depth overview of disaster management systems, methods, obstacles, and potential future paths. Specifically, it focuses on flood control, a significant and recurrent category of natural disasters. The analysis begins by exploring various types of natural catastrophes, including earthquakes, wildfires, and floods. It then delves into the different domains that collectively contribute to effective flood management. These domains encompass cutting-edge technologies such as big data analysis and cloud computing, providing scalable and reliable infrastructure for data storage, processing, and analysis. The study investigates the potential of the Internet of Things and sensor networks to gather real-time data from flood-prone areas, enhancing situational awareness and enabling prompt actions. Model-driven engineering is examined for its utility in developing and modeling flood scenarios, aiding in preparation and response planning. This study includes the Google Earth engine (GEE) and examines previous studies involving GEE. Moreover, we discuss remote sensing; remote sensing is undoubtedly a valuable tool for disaster management, and offers geographical data in various situations. We explore the application of Geographical Information System (GIS) and Spatial Data Management for visualizing and analyzing spatial data and facilitating informed decision-making and resource allocation during floods. In the final section, the focus shifts to the utilization of machine learning and data analytics in flood management. These methodologies offer predictive models and data-driven insights, enhancing early warning systems, risk assessment, and mitigation strategies. Through this in-depth analysis, the significance of incorporating these spheres into flood control procedures is highlighted, with the aim of improving disaster management techniques and enhancing resilience in flood-prone regions. The paper addresses existing challenges and provides future research directions, ultimately striving for a clearer and more coherent representation of disaster management techniques.

Keywords: disaster management; natural disasters; floods; wildfire; earthquake; ecosystem

check for updates

Citation: Khan, S.M.; Shafi, I.; Butt, W.H.; Diez, I.d.l.T.; Flores, M.A.L.; Galán, J.C.; Ashraf, I. A Systematic Review of Disaster Management Systems: Approaches, Challenges, and Future Directions. *Land* **2023**, *12*, 1514. https://doi.org/10.3390/land12081514

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1. Introduction

Natural catastrophes such as earthquakes, tsunamis, floods, forest fires, plane crashes, and viruses are becoming more common, posing major challenges not just for the public

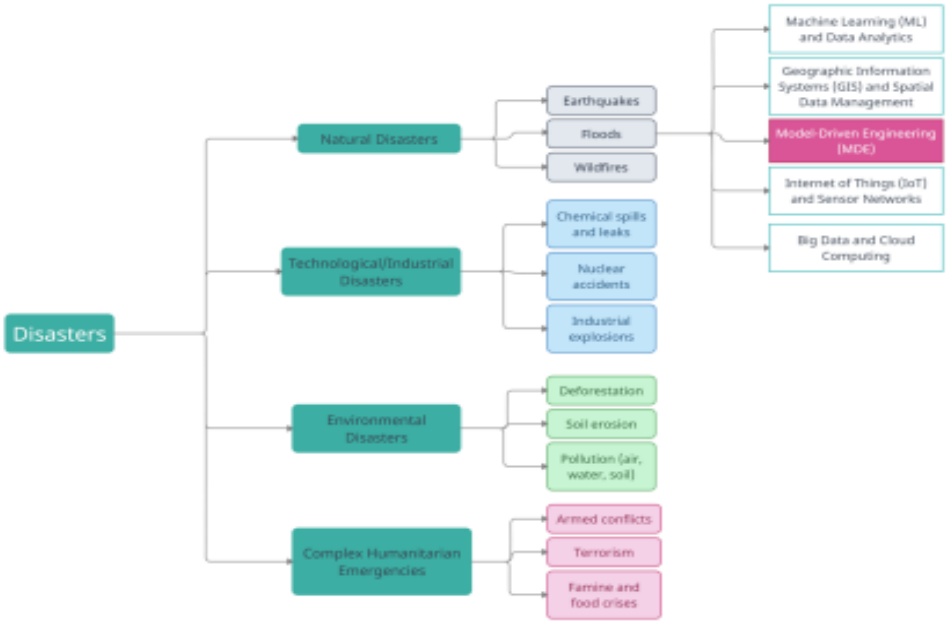


Figure 1. Block diagram showing the workflow followed in this review.

2. Research Methodology

To carry out the study process known as a systematic review for the topic of disaster management systems, we used the methodology shown in Figure 2 for data collection and analysis.

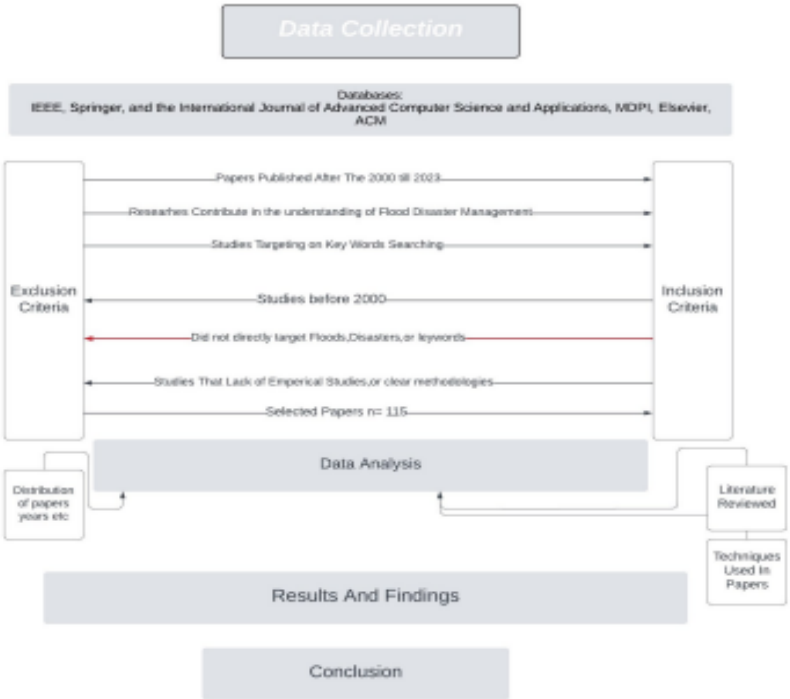
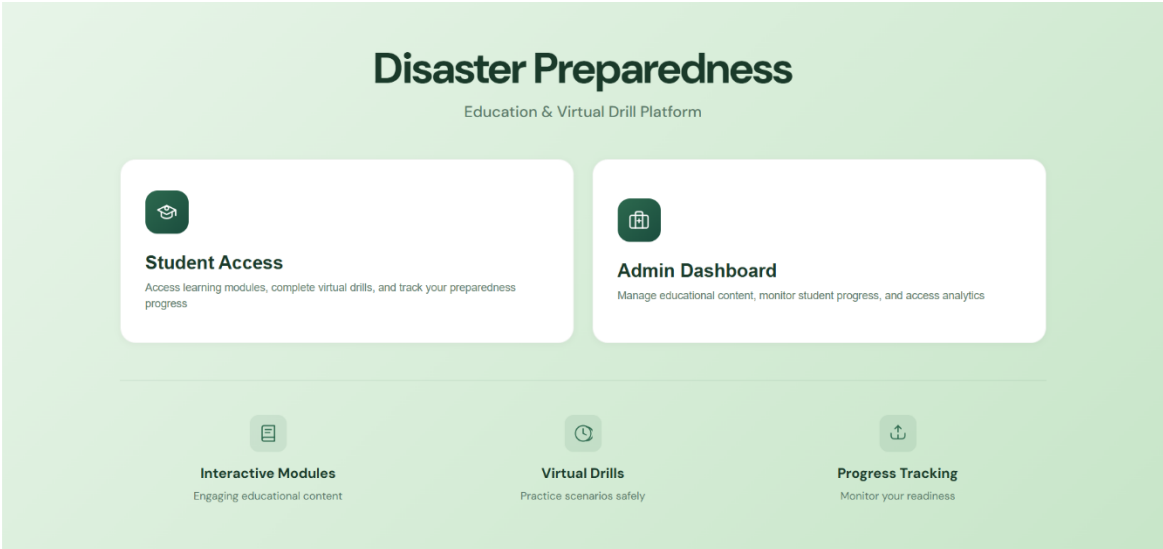
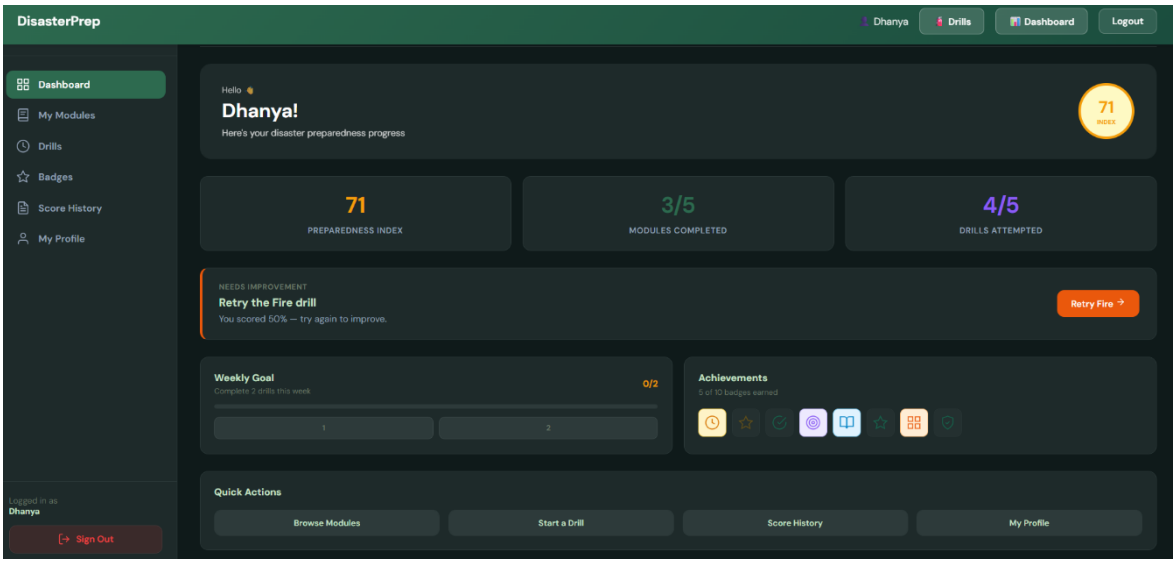


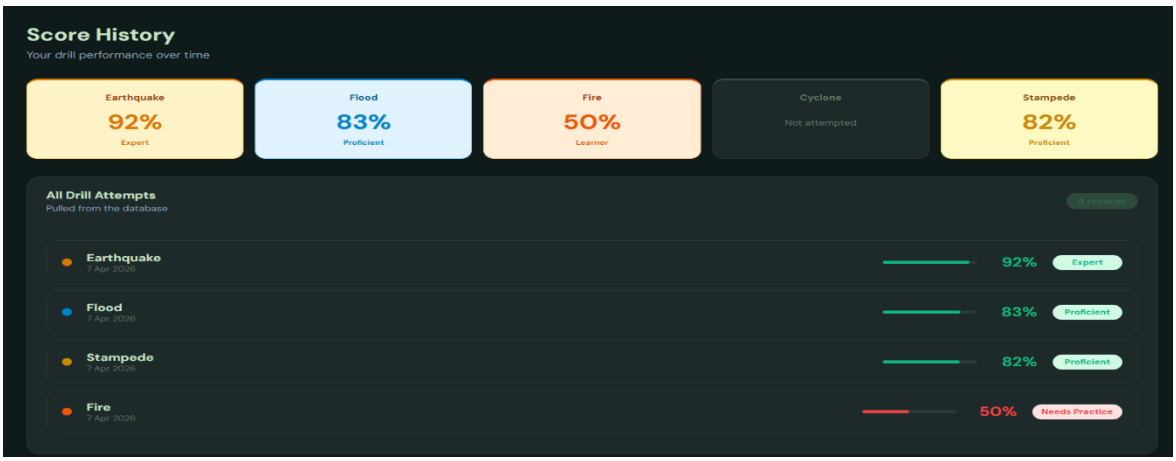
Figure 2. Methodology adopted for the literature review.

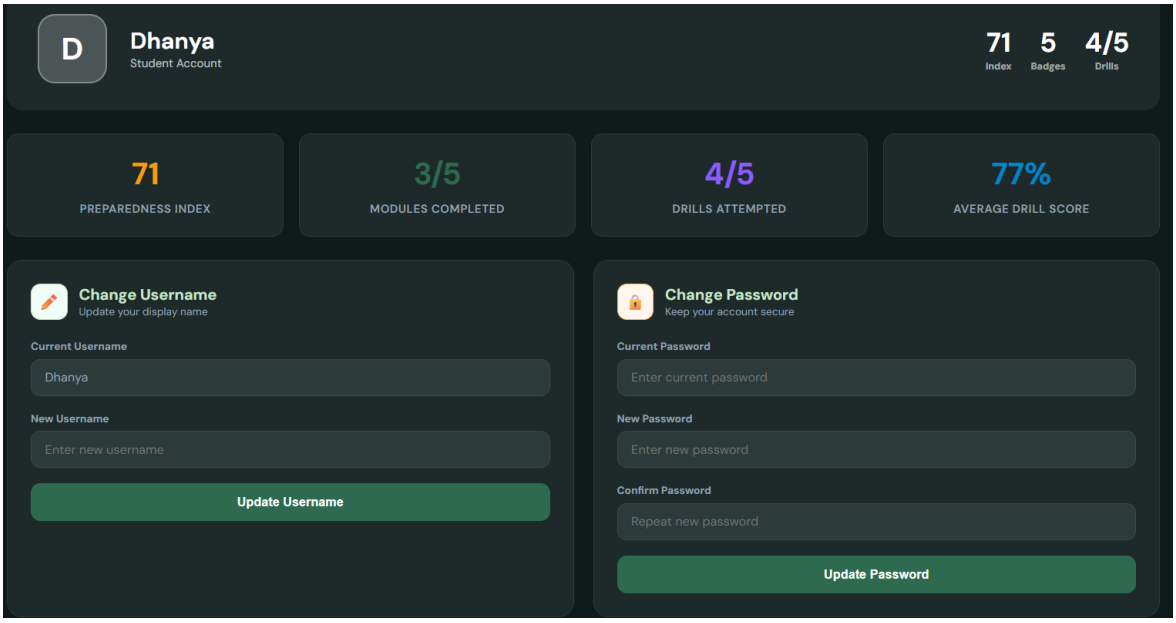
APPENDICES

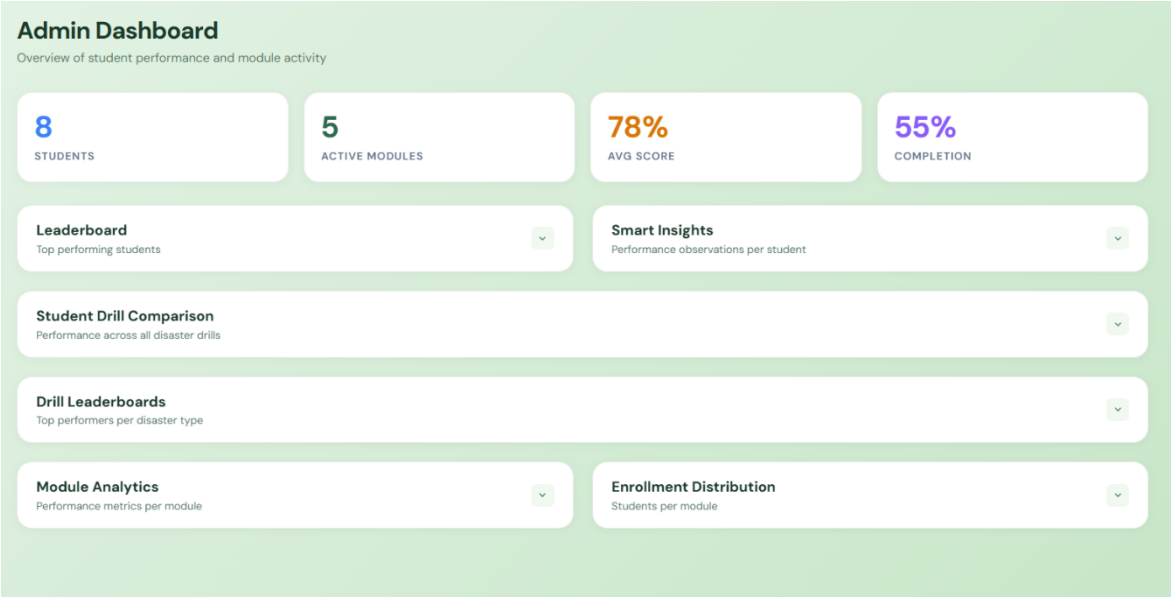
Few images of project:

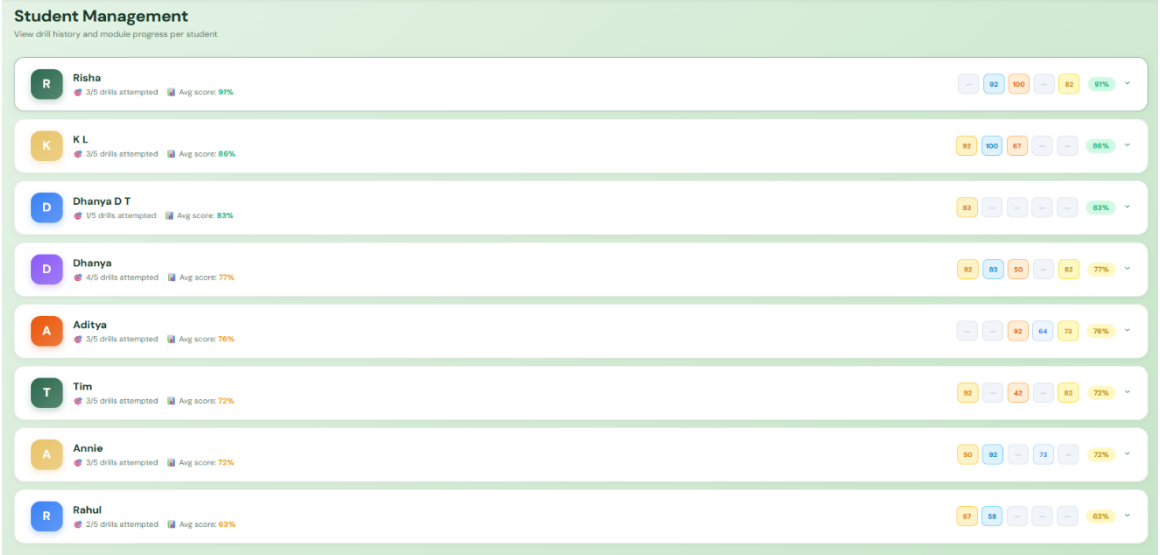


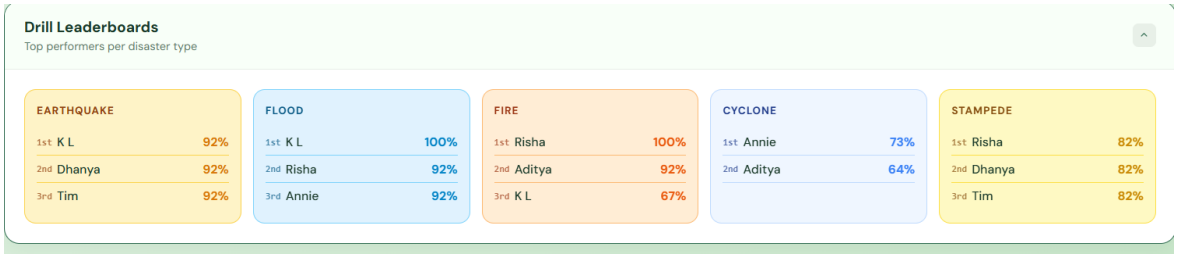


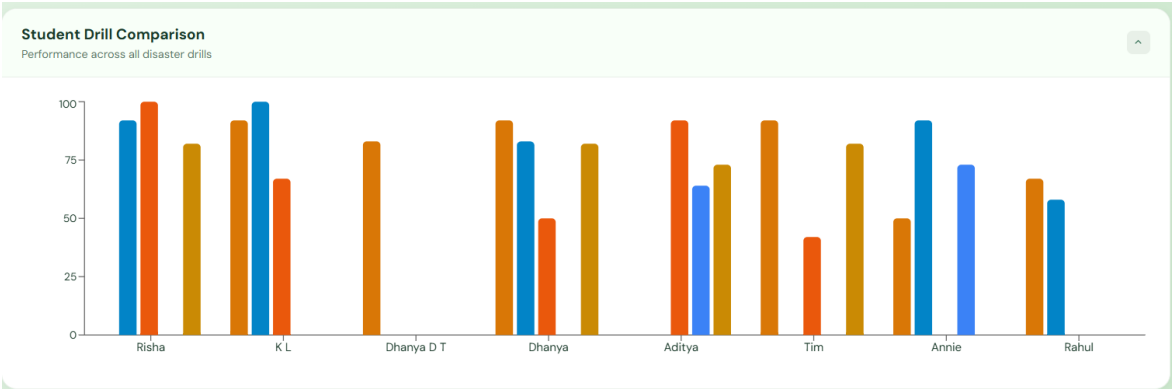


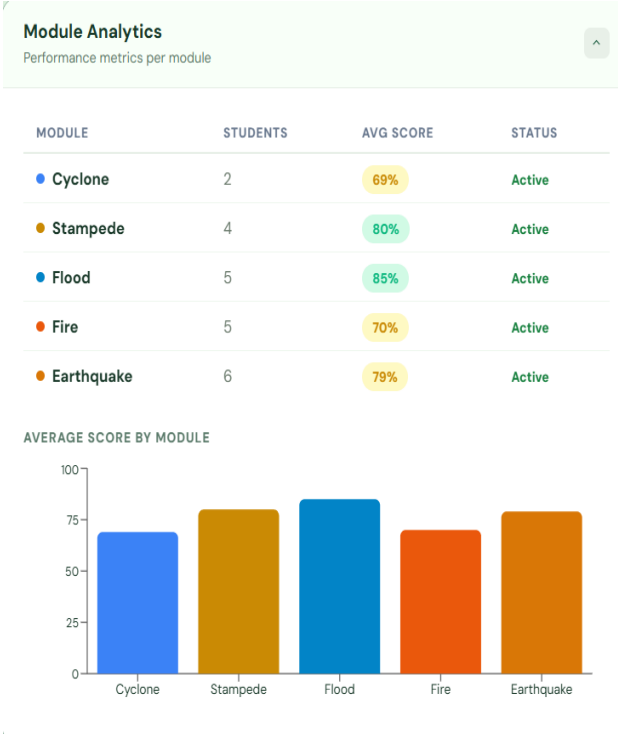

D
Dhanya
Student Account
71
Index
5
Badges
4/5
Drills
71
PREPAREDNESS INDEX
3/5
MODULES COMPLETED
4/5
DRILLS ATTEMPTED
77%
AVERAGE DRILL SCORE
Change Username
Update your display name
Current Username
Dhanya
New Username
Enter new username
Update Username
Change Password
Keep your account secure
Current Password
Enter current password
New Password
Enter new password
Confirm Password
Repeat new password
Update Password

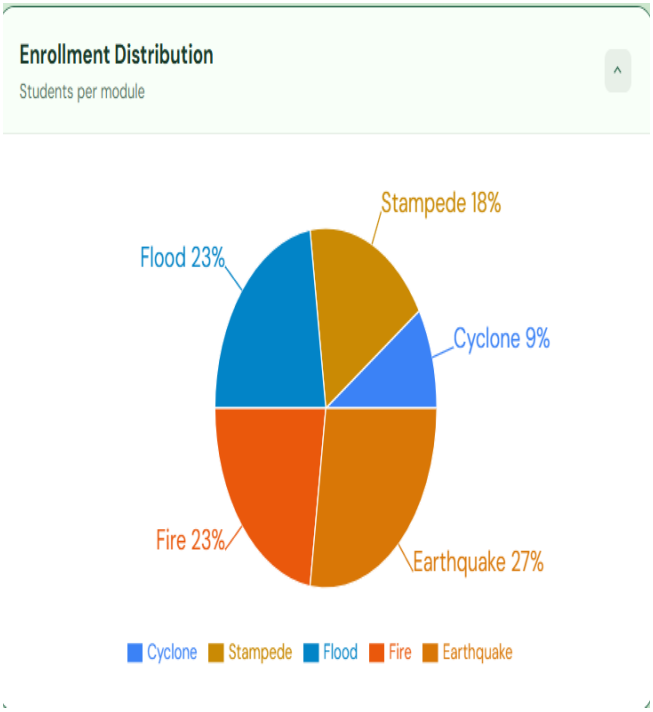

Admin Dashboard
Overview of student performance and module activity
8
STUDENTS
5
ACTIVE MODULES
78%
AVG SCORE
55%
COMPLETION
Leaderboard
Top performing students
Smart Insights
Performance observations per student
Student Drill Comparison
Performance across all disaster drills
Drill Leaderboards
Top performers per disaster type
Module Analytics
Performance metrics per module
Enrollment Distribution
Students per module


Student Management
View drill history and module progress per student
Risha
K L
Dhanya D T
Dhanya
Aditya
Tim
Annie
Rahul


Drill Leaderboards
Top performers per disaster type
EARTHQUAKE
1st K L 92%
2nd Dhanya 92%
3rd Tim 92%
FLOOD
1st K L 100%
2nd Risha 92%
3rd Annie 92%
FIRE
1st Risha 100%
2nd Aditya 92%
3rd K L 67%
CYCLONE
1st Annie 73%
2nd Aditya 64%
STAMPEDE
1st Risha 82%
2nd Dhanya 82%
3rd Tim 82%


Student Drill Comparison
Performance across all disaster drills
100
75
50
25
0
Risha
K L
Dhanya D T
Dhanya
Aditya
Tim
Annie
Rahul


Module Analytics
Performance metrics per module
MODULE STUDENTS AVG SCORE STATUS
Cyclone 2 69% Active
Stampede 4 80% Active
Flood 5 85% Active
Fire 5 70% Active
Earthquake 6 79% Active
AVERAGE SCORE BY MODULE
100
75
50
25
0
Cyclone
Stampede
Flood
Fire
Earthquake


Enrollment Distribution
Students per module
Stampede 18%
Flood 23%
Cyclone 9%
Fire 23%
Earthquake 27%
Cyclone
Stampede
Flood
Fire
Earthquake

Submission Summary

Print Email

Conference Name	2nd International Conference on Emerging Technologies in Computer Science for Interdisciplinary Applications
Track Name	System Modeling and Computing
Paper ID	12
Paper Title	Disaster Preparedness and Response Education System for Schools and Colleges
Abstract	Natural disasters such as earthquakes, floods, cyclones, and fires pose serious threats to human life and infrastructure. One of the major challenges during such situations is the lack of awareness and improper decision-making among individuals. Traditional disaster education methods are mostly theoretical and fail to prepare individuals for real-life emergencies. This paper presents an interactive disaster management training system based on simulation-driven learning. The system is implemented using a full-stack architecture and provides modules, drill-based scenarios, and performance evaluation through a preparedness index. It enhances user engagement and improves disaster awareness through realistic training approaches. The proposed system offers an effective and scalable solution for disaster preparedness education in schools and colleges.
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Authors	**Dhanya D T** (Presidency University) <dhanyadt6@gmail.com>
Submission Files	CC9_RESEARCHPAPER.pdf (353.8 Kb, 2026-04-21 15:52:34 GMT+5:30)

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Paper ID: 12

Paper Title: Disaster Preparedness and Response Education System for Schools and Colleges

Abstract:
Natural disasters such as earthquakes, floods, cyclones, and fires pose serious threats to human life and infrastructure. One of the major challenges during such situations is the lack of awareness and improper decision-making among individuals. Traditional disaster education methods are mostly theoretical and fail to prepare individuals for real-life emergencies.

This paper presents an interactive disaster management training system based on simulation-driven learning. The system is implemented using a full-stack architecture and provides modules, drill-based scenarios, and performance evaluation through a preparedness index. It enhances user engagement and improves disaster awareness through realistic training approaches.

The proposed system offers an effective and scalable solution for disaster preparedness education in schools and colleges.

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